Pruebas a la web api usando las herramientas de postman

Para crear pruebas a la web api con postman, te recomendamos que antes de comenzar, leas los siguientes artículos:

https://learning.postman.com/docs/writing-scripts/script-references/test-examples/

https://learning.postman.com/docs/writing-scripts/script-references/variables-list/

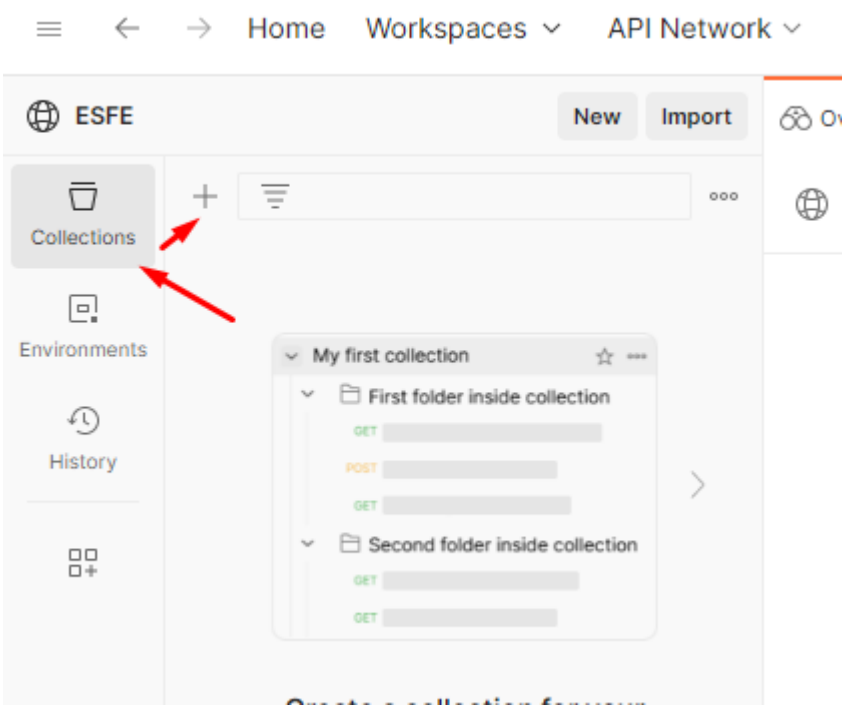
https://www.postman.com/templates/da8ceebc-d4da-478c-ab8a-37634faefc74/Mock-data-generation

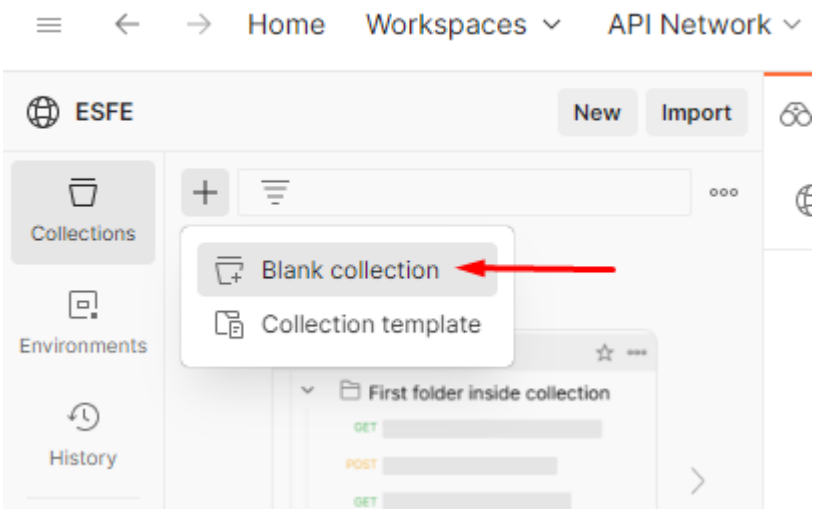
Para enseñar los test en Postman a realizar a la web api .net core vamos utilizar un proyecto que construimos en las guías anteriores.

https://docs.google.com/document/d/190HZwykTRjKLCclve30AV_WhSssO5HZyfFoe7dtjoK0/edit?usp=sharing

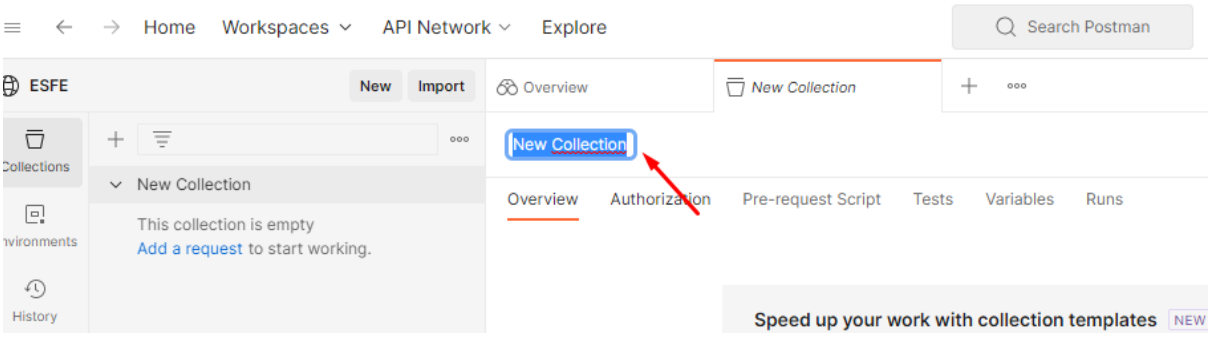
https://github.com/ESFEAlumnos2023/webapimvcblazor

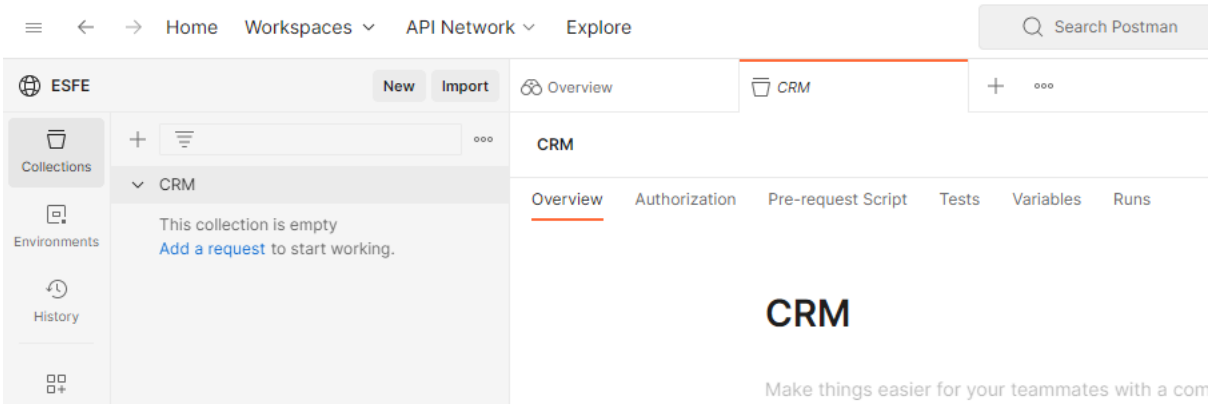
Vamos a comenzar creando una collection en Postman para agrupar todos nuestros endpoints del proyecto CRM.



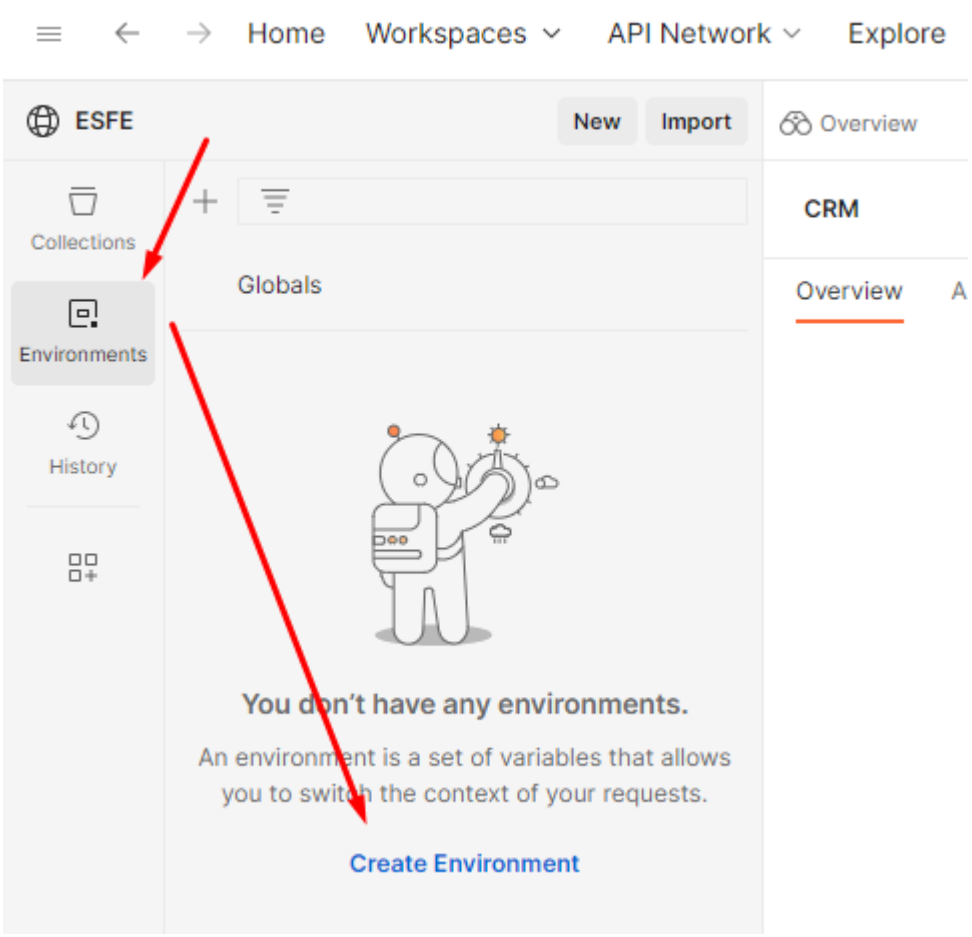


Vamos a cambiar el nombre a 'CRM'.

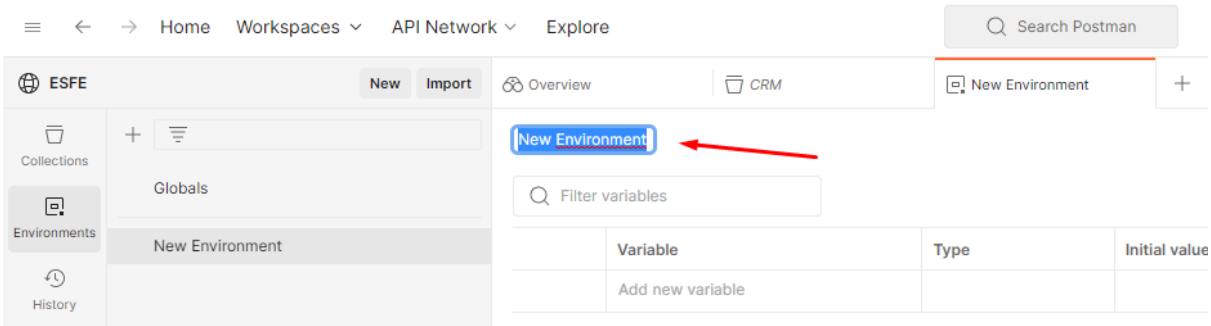




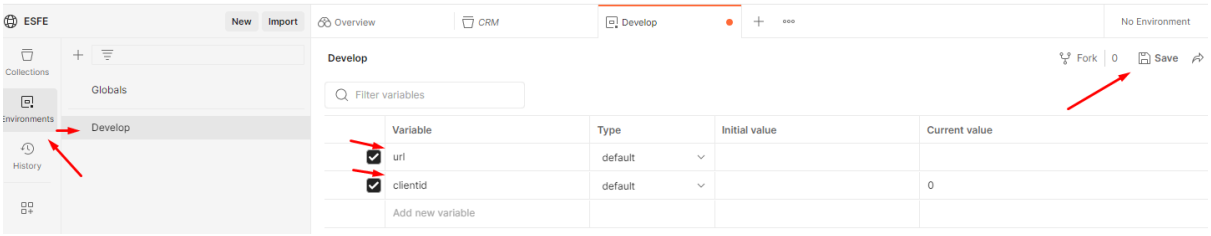
Lo siguiente que haremos es crear nuestras variables globales, las cuales nos servirán para tener configuraciones generales al realizar pruebas en una API web con Postman.



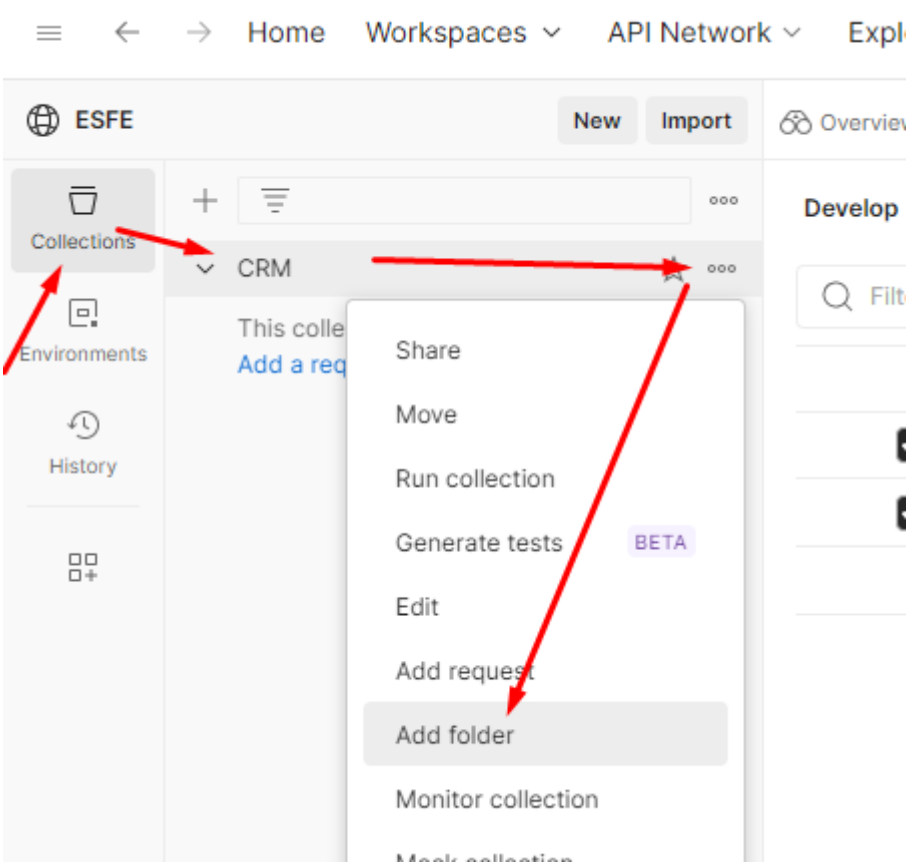
Le agregaremos el nombre 'Develop'.

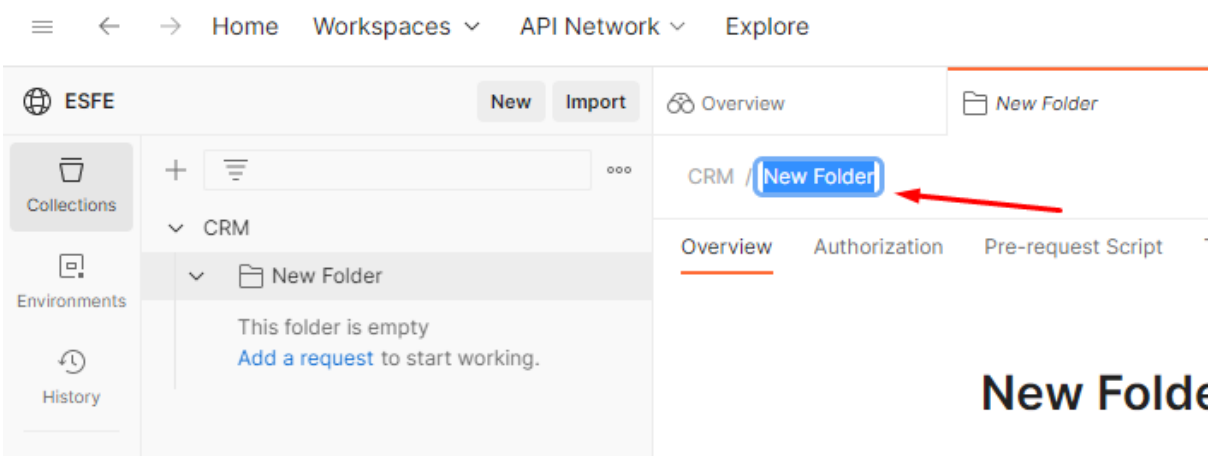


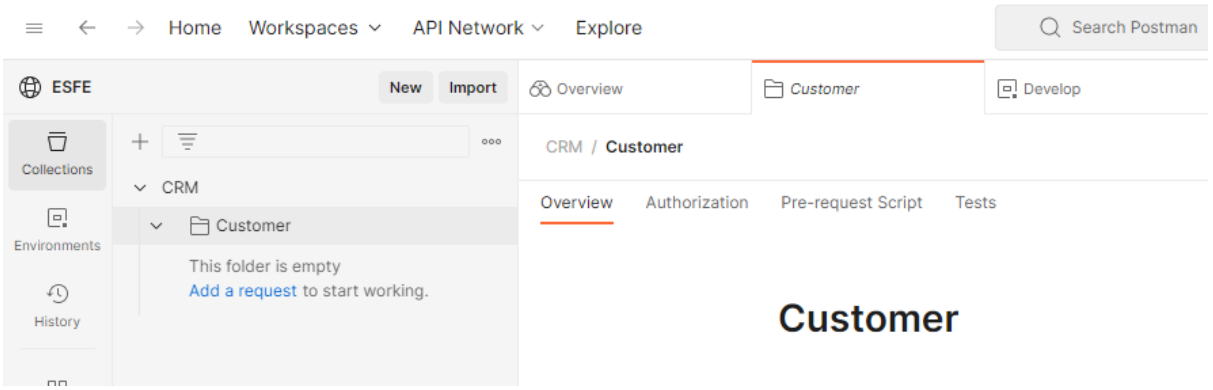
Lo siguiente es crear las variables globales:
url: Donde se guardará la ruta principal de nuestra API web.
clientId: Donde se almacenará el ID del cliente que se utilizará en las pruebas.



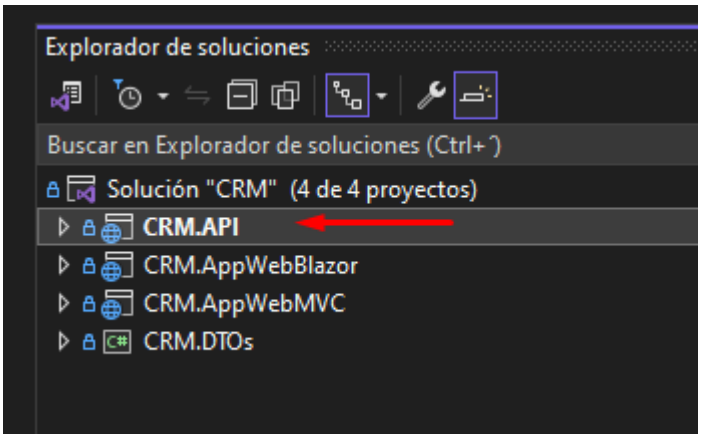
Lo siguiente es crear una carpeta con el nombre 'Customer' para agrupar todos los endpoints relacionados con el proceso de manipulación de información de los clientes.

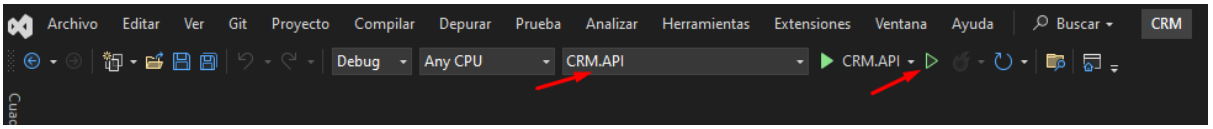




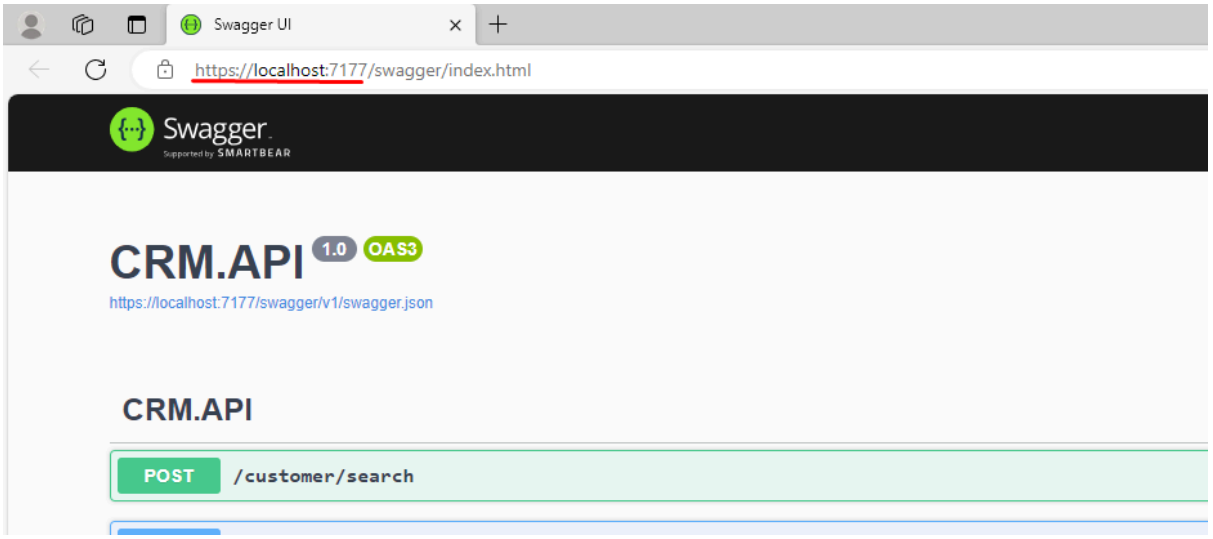


Ahora, vamos a iniciar el proyecto de una API web en Visual Studio 2022.

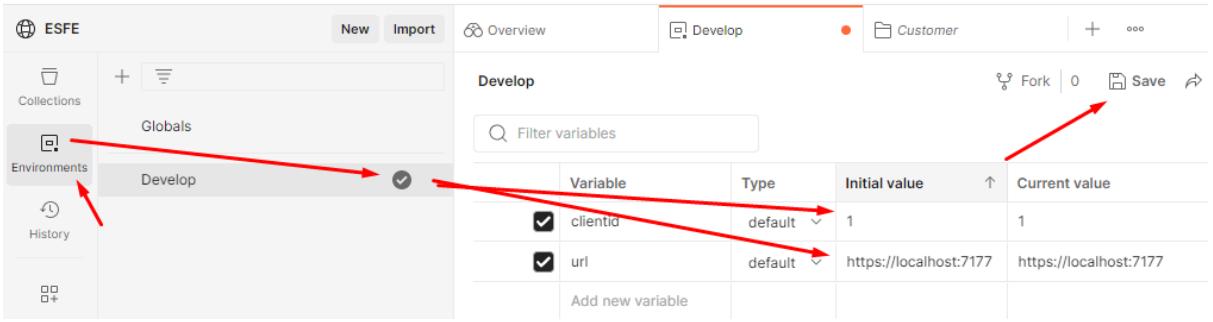




Copiamos la URL donde nuestra API web está en funcionamiento.



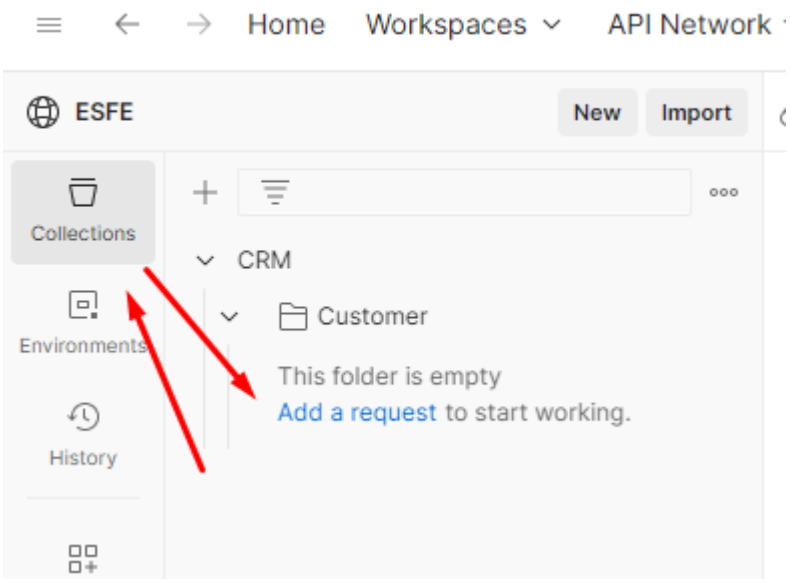
Pegamos esa URL en la variable global de Postman y establecemos la variable clientId con el valor 1.

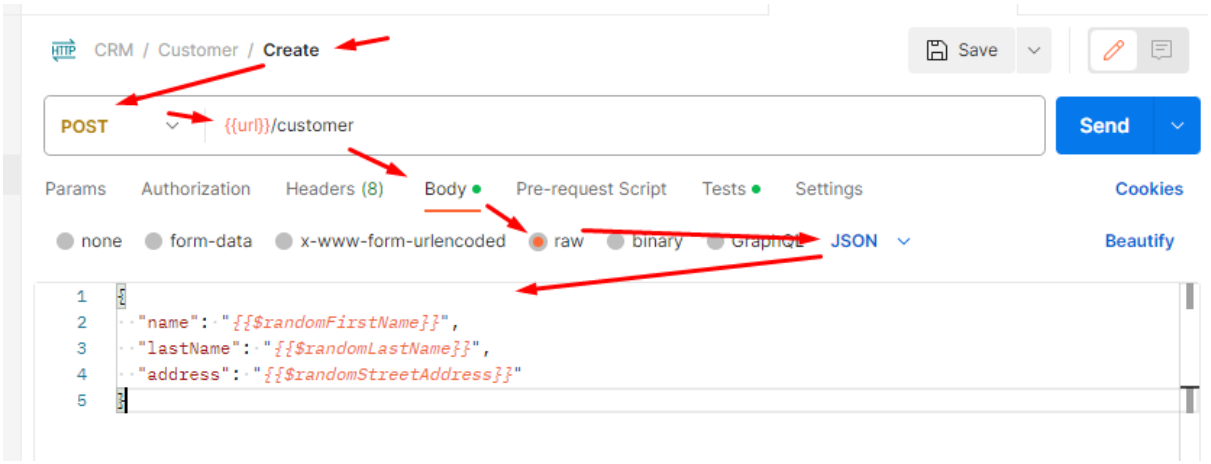


Lo siguiente será crear los endpoints para crear, modificar, eliminar, buscar y obtener por ID de cliente.

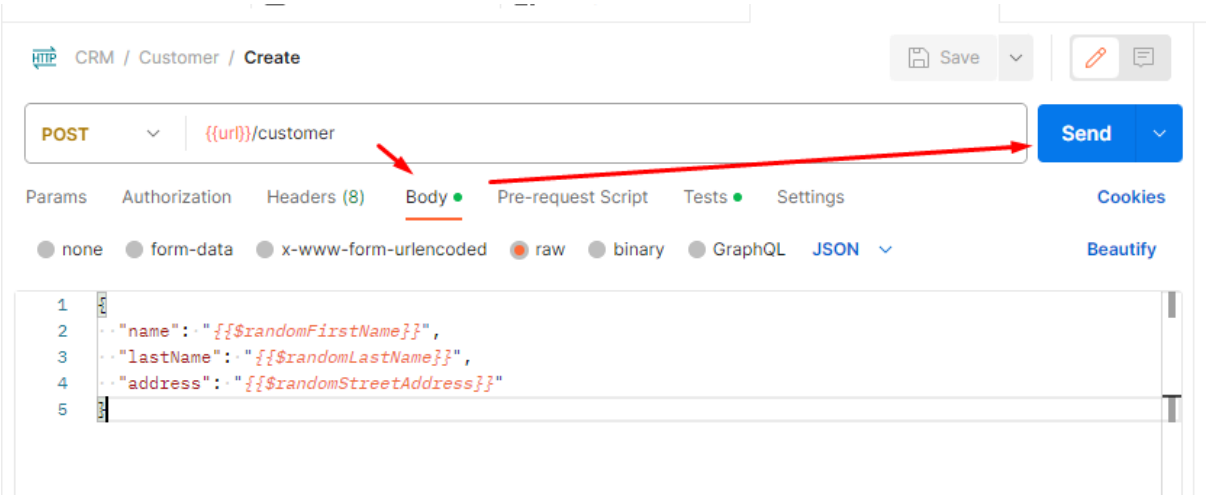
En postman hacemos uso de {{}} para poder llamar variables globales o obtener valores aleatorios {{$}}

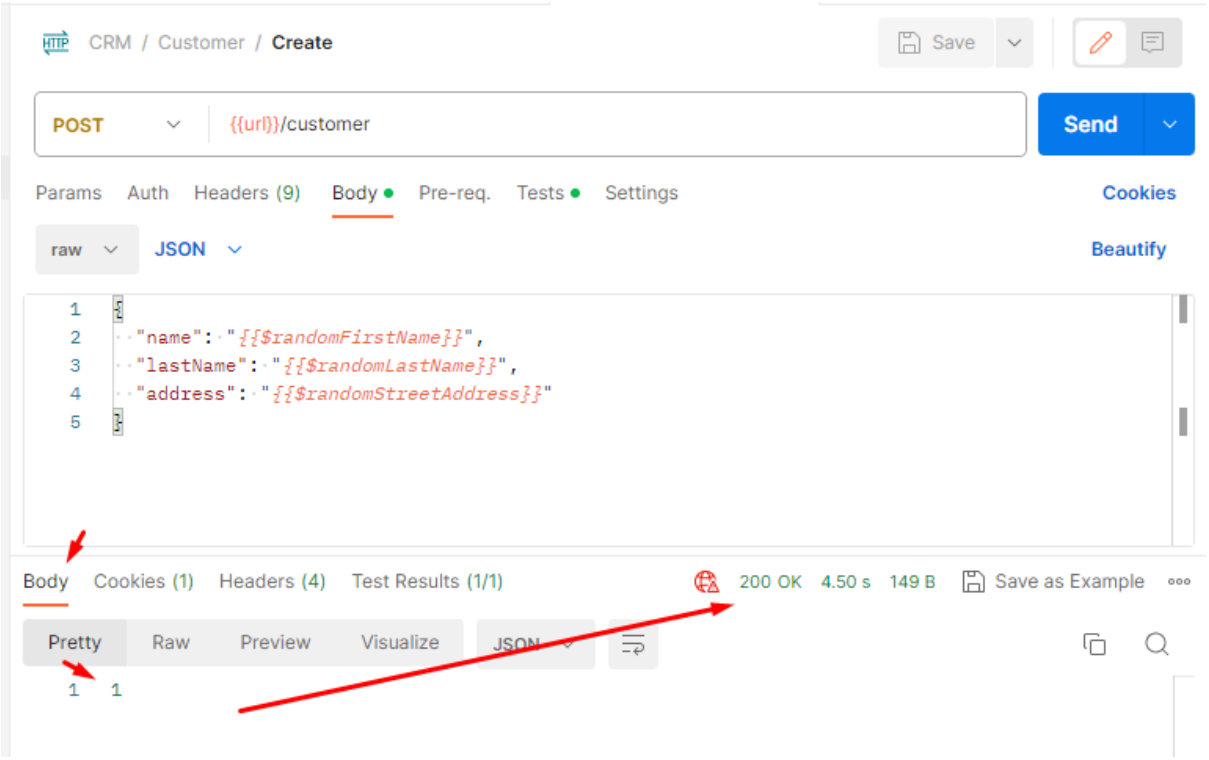
Comenzaremos creando endpoints para crear clientes.

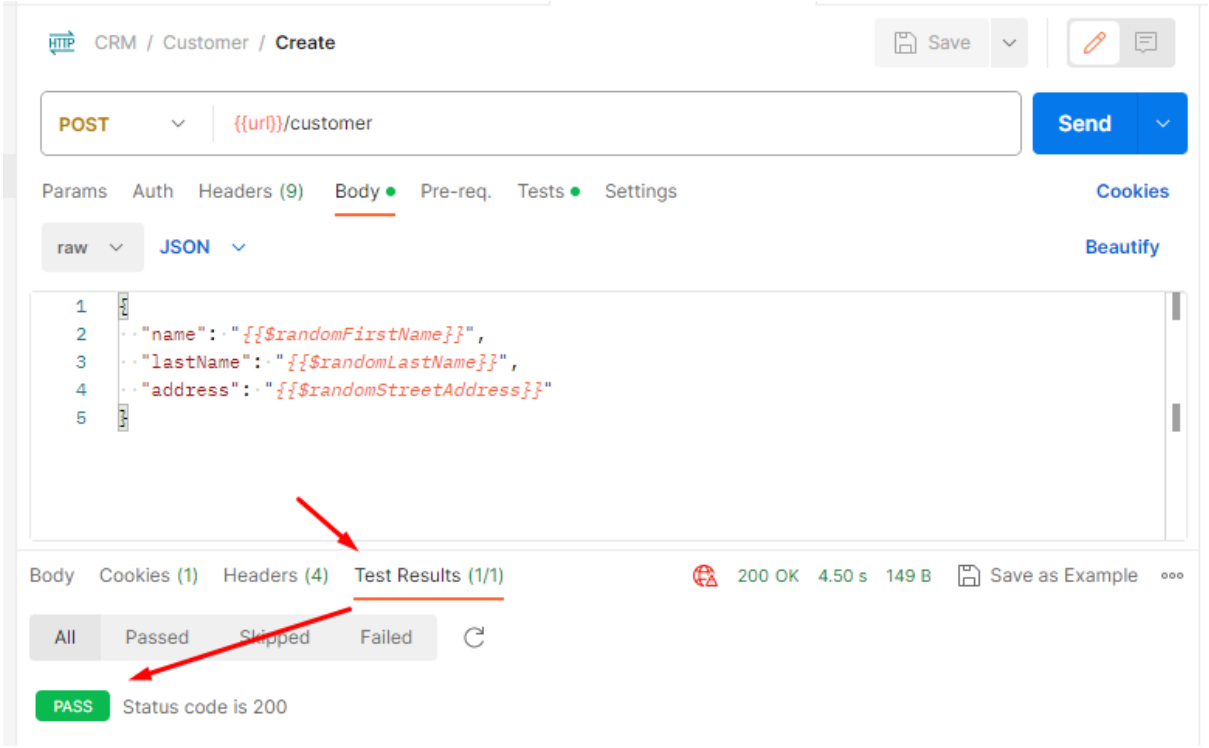

Home
Workspaces
API Network
ESFE
New
Import
Collections
Environments
History
CRM
Customer
This folder is empty
Add a request to start working.

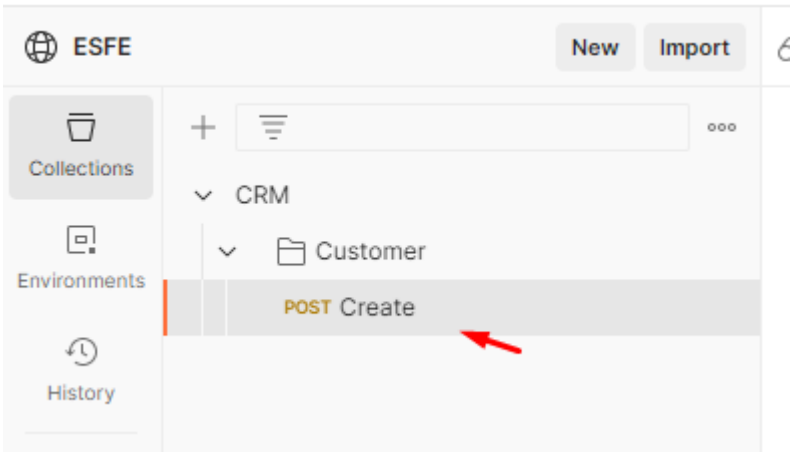

CRM / Customer / Create
Save
POST
{{url}}/customer
Send
Params
Authorization
Headers (8)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"name": "{{$randomFirstName}}",
"lastName": "{{$randomLastName}}",
"address": "{{$randomStreetAddress}}"
}


CRM / Customer / Create
Save
POST
{{url}}/customer
Send
Params
Authorization
Headers (8)
Body
Pre-request Script
Tests
Settings
Cookies
pm.test("Status code is 200", function () {
pm.response.to.have.status(200);
});
Write scripts with Postbot
Test scripts are written in JavaScript, and are run after the response is received.

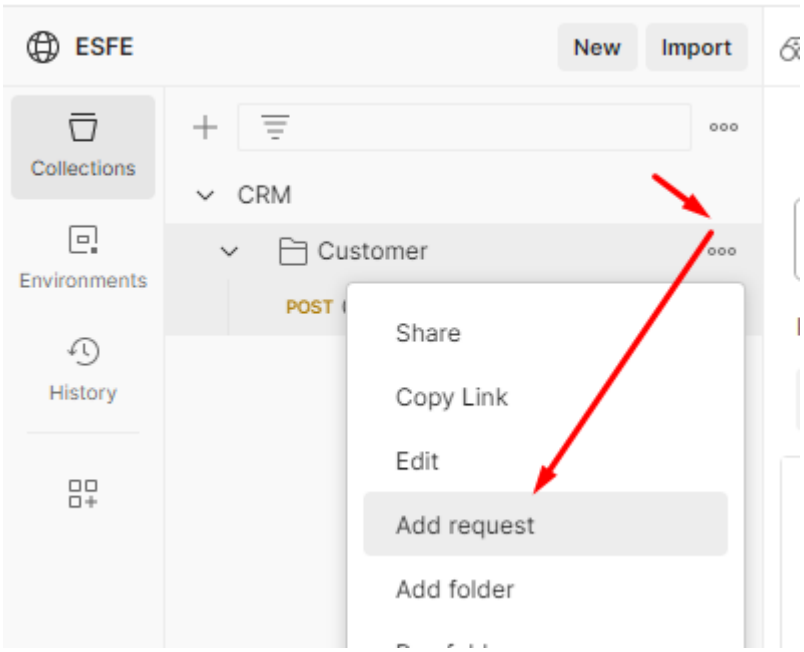

CRM / Customer / Create
Save
POST
{{url}}/customer
Send
Params
Authorization
Headers (8)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"name": "{{$randomFirstName}}",
"lastName": "{{$randomLastName}}",
"address": "{{$randomStreetAddress}}"
}

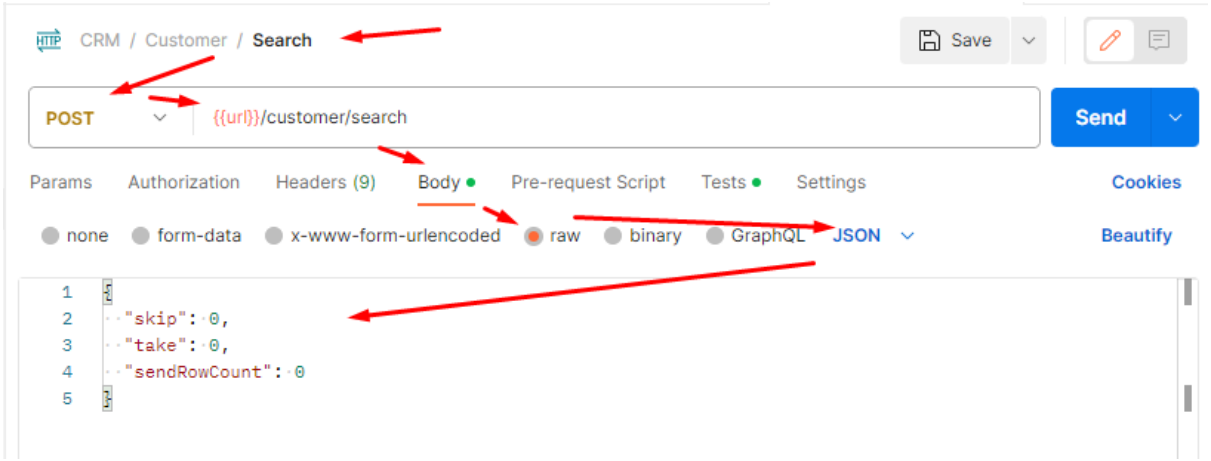

CRM / Customer / Create
Save
POST
{{url}}/customer
Send
Params
Auth
Headers (9)
Body
Pre-req.
Tests
Settings
Cookies
raw
JSON
Beautify
{
"name": "{{$randomFirstName}}",
"lastName": "{{$randomLastName}}",
"address": "{{$randomStreetAddress}}"
}
Body
Cookies (1)
Headers (4)
Test Results (1/1)
200 OK
4.50 s
149 B
Save as Example
Pretty
Raw
Preview
Visualize
JSON
1

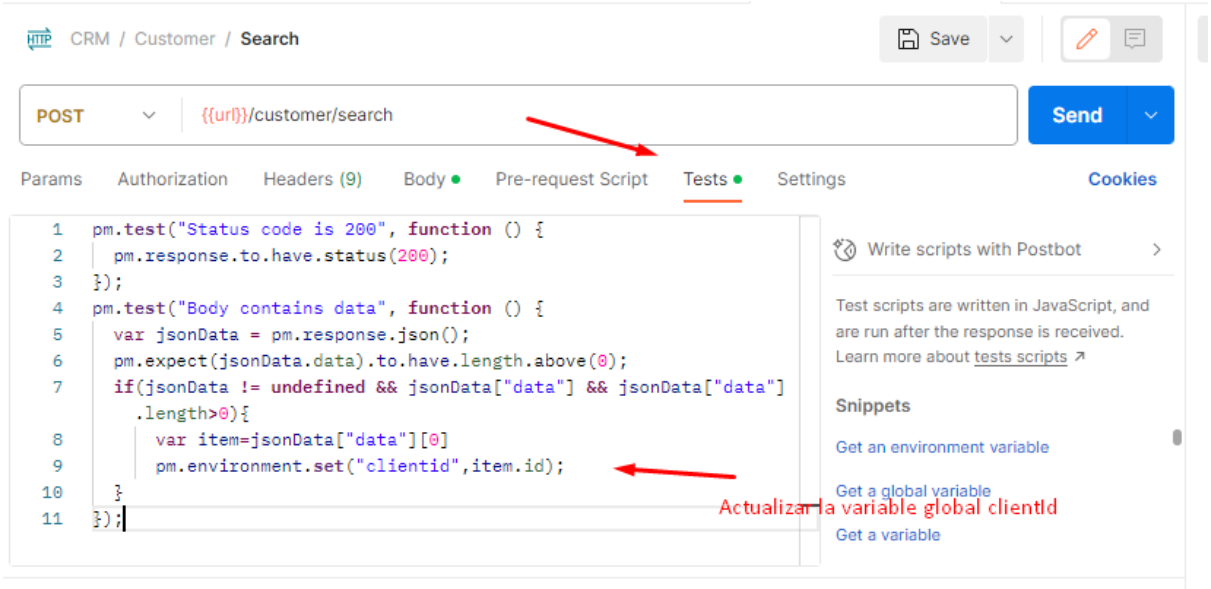


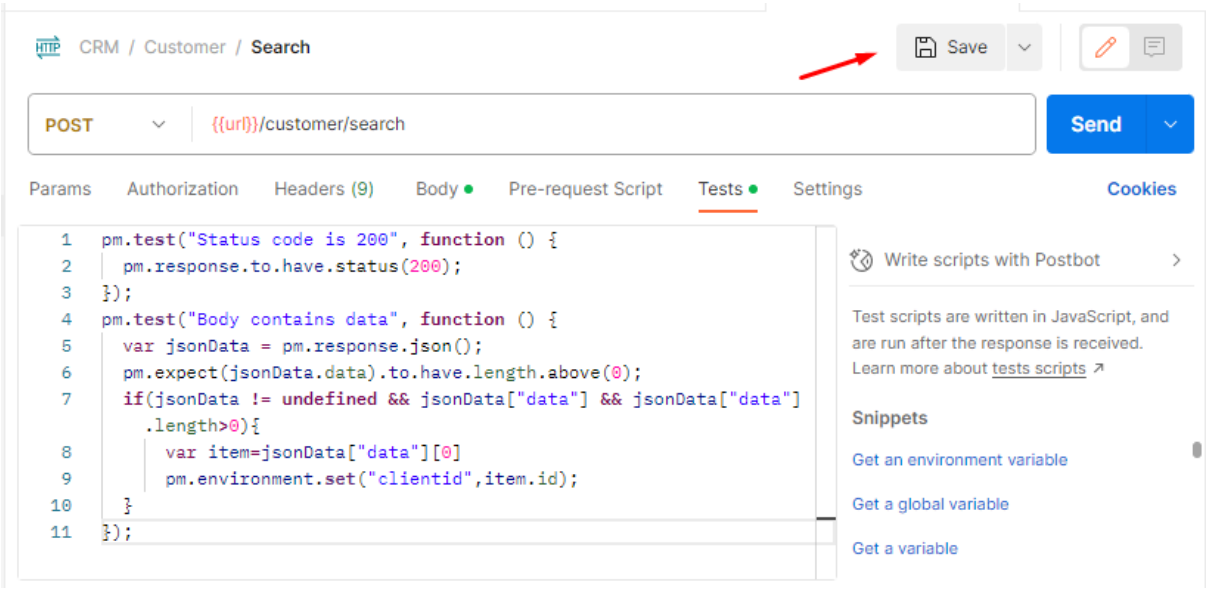


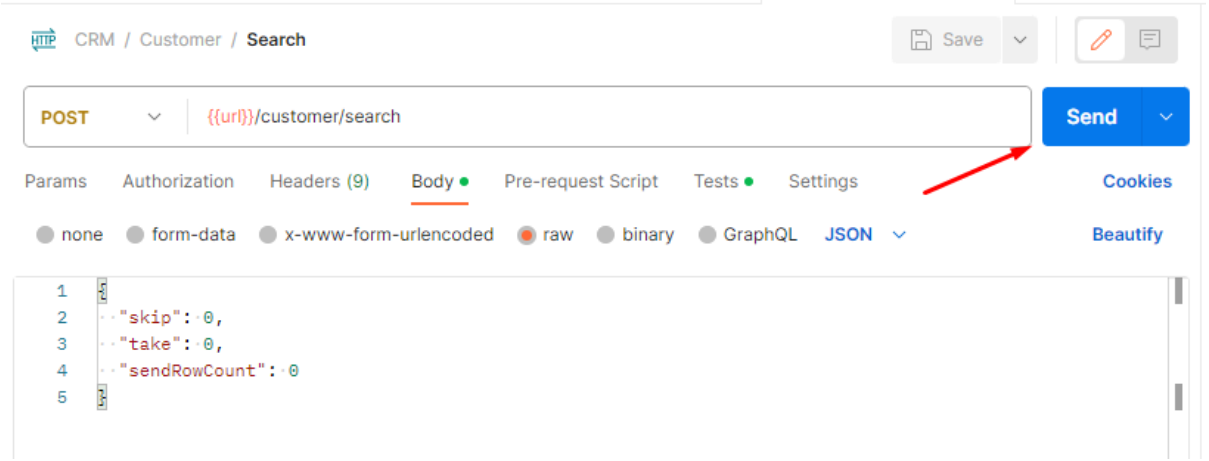
Ahora vamos a crear el endpoint para buscar clientes.


ESFE
New
Import
Collections
Environments
History
CRM
Customer
POST
Share
Copy Link
Edit
Add request
Add folder

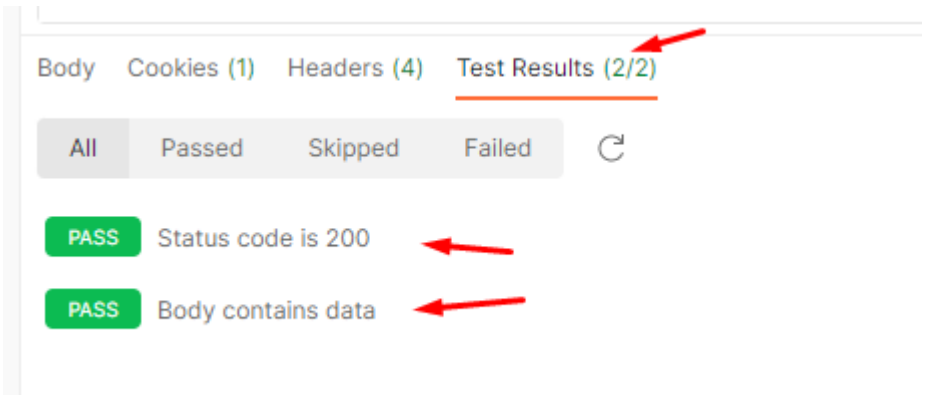

CRM / Customer / Search
Save
POST
{{url}}/customer/search
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"skip": 0,
"take": 0,
"sendRowCount": 0
}

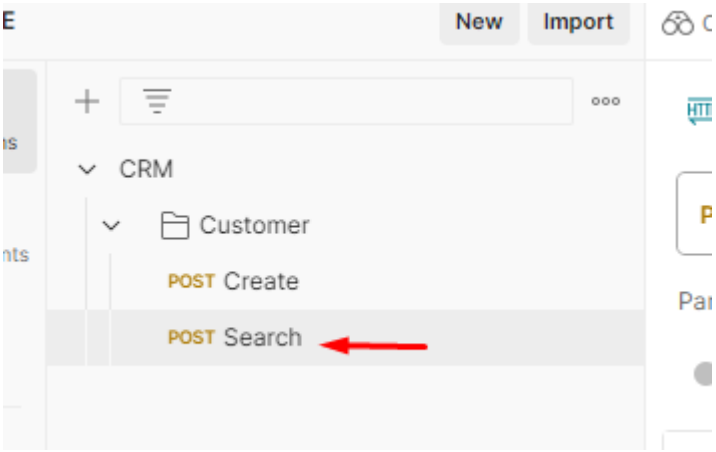

CRM / Customer / Search
Save
POST
{{url}}/customer/search
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
pm.test("Status code is 200", function () {
pm.response.to.have.status(200);
});
pm.test("Body contains data", function () {
var jsonData = pm.response.json();
pm.expect(jsonData.data).to.have.length.above(0);
if(jsonData != undefined && jsonData["data"] && jsonData["data"].length>0){
var item=jsonData["data"][0]
pm.environment.set("clientid",item.id);
}
});
Write scripts with Postbot
Test scripts are written in JavaScript, and are run after the response is received.
Learn more about tests scripts
Snippets
Get an environment variable
Get a global variable
Get a variable
Actualizar la variable global clientId


CRM / Customer / Search
Save
POST
{{url}}/customer/search
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
pm.test("Status code is 200", function () {
pm.response.to.have.status(200);
});
pm.test("Body contains data", function () {
var jsonData = pm.response.json();
pm.expect(jsonData.data).to.have.length.above(0);
if(jsonData != undefined && jsonData["data"] && jsonData["data"].length>0){
var item=jsonData["data"][0]
pm.environment.set("clientid",item.id);
}
});
Write scripts with Postbot
Test scripts are written in JavaScript, and are run after the response is received.
Learn more about tests scripts
Snippets
Get an environment variable
Get a global variable
Get a variable

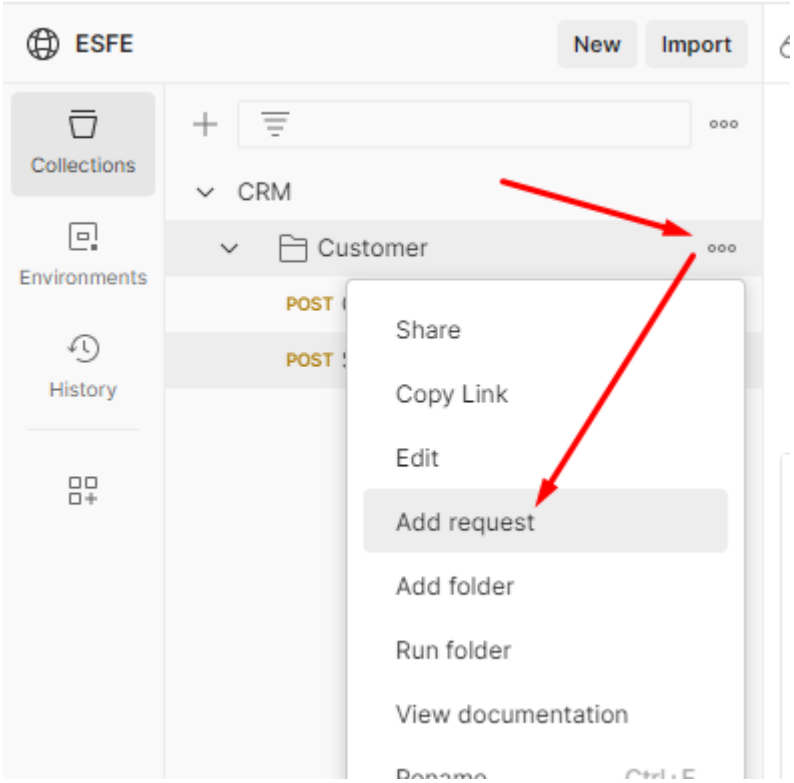

CRM / Customer / Search
Save
POST
{{url}}/customer/search
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"skip": 0,
"take": 0,
"sendRowCount": 0
}

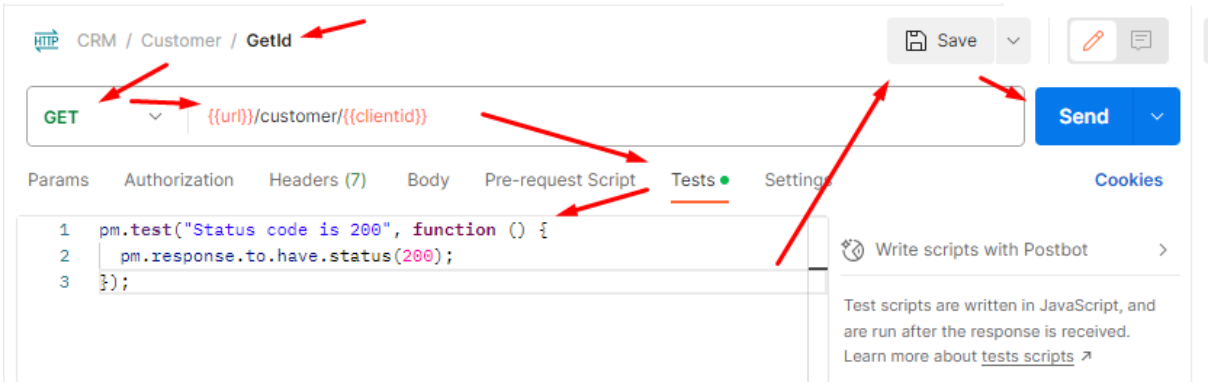

Body
Cookies (1)
Headers (4)
Test Results (2/2)
200 OK
622 ms
869 B
Save as Example
Pretty
Raw
Preview
Visualize
JSON
{
"countRow": 0,
"data": [
{
"id": 3002,
"name": "Patrick",
"lastName": "Thiel",
"address": "7975 Effertz Haven"
},
{

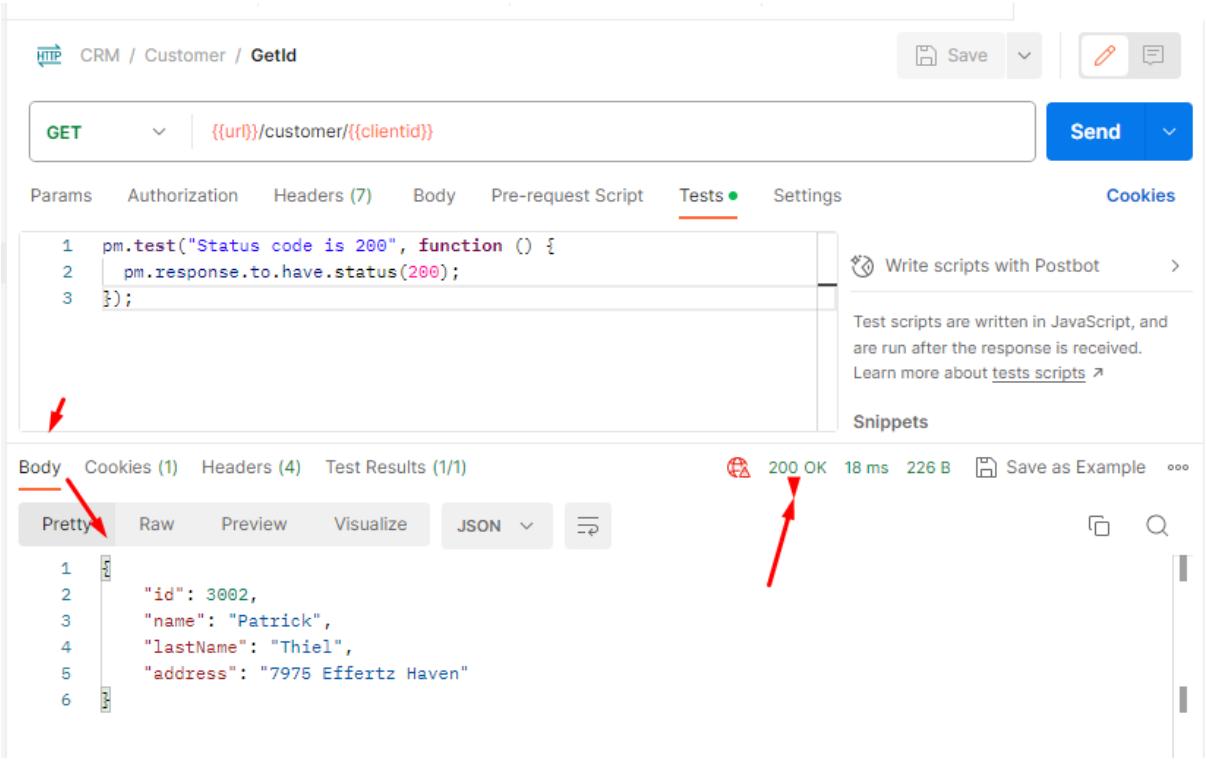


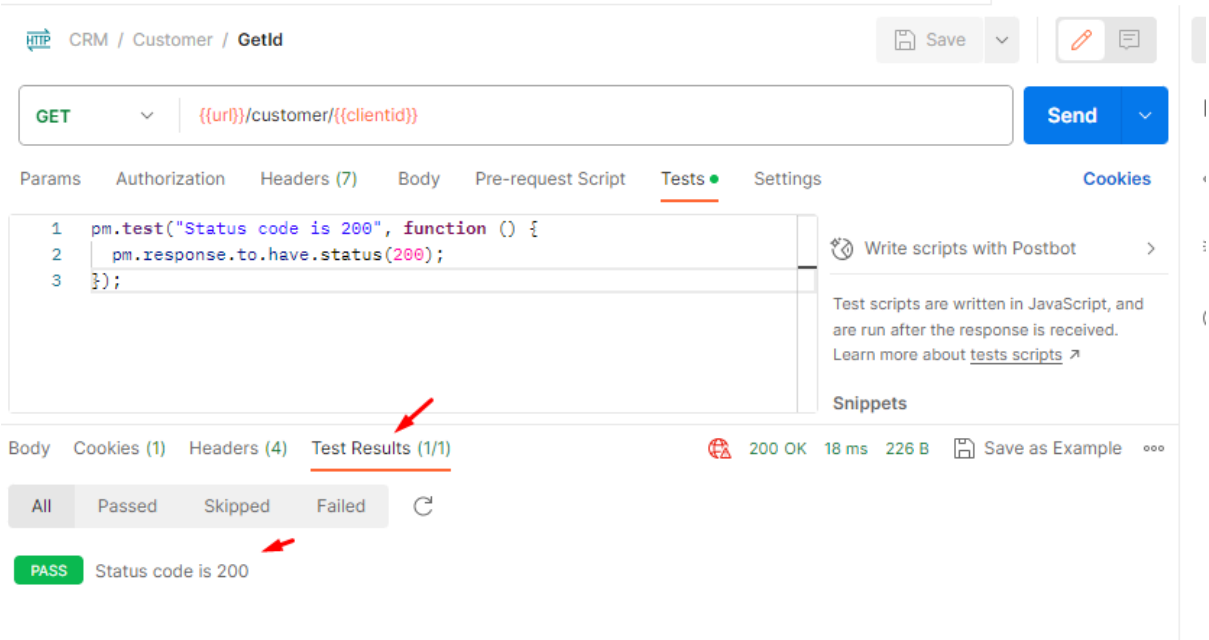


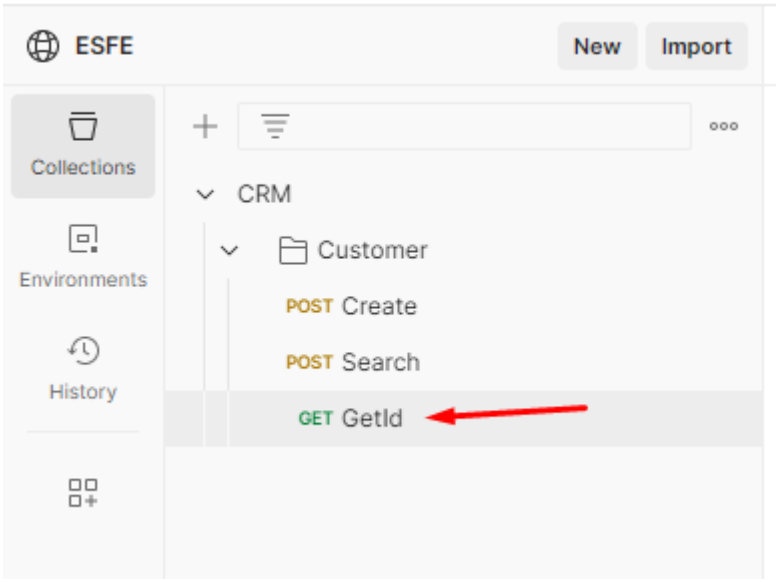
Ahora vamos a crear el endpoint para obtener por id.


ESFE
New
Import
Collections
Environments
History
CRM
Customer
POST
POST
Share
Copy Link
Edit
Add request
Add folder
Run folder
View documentation

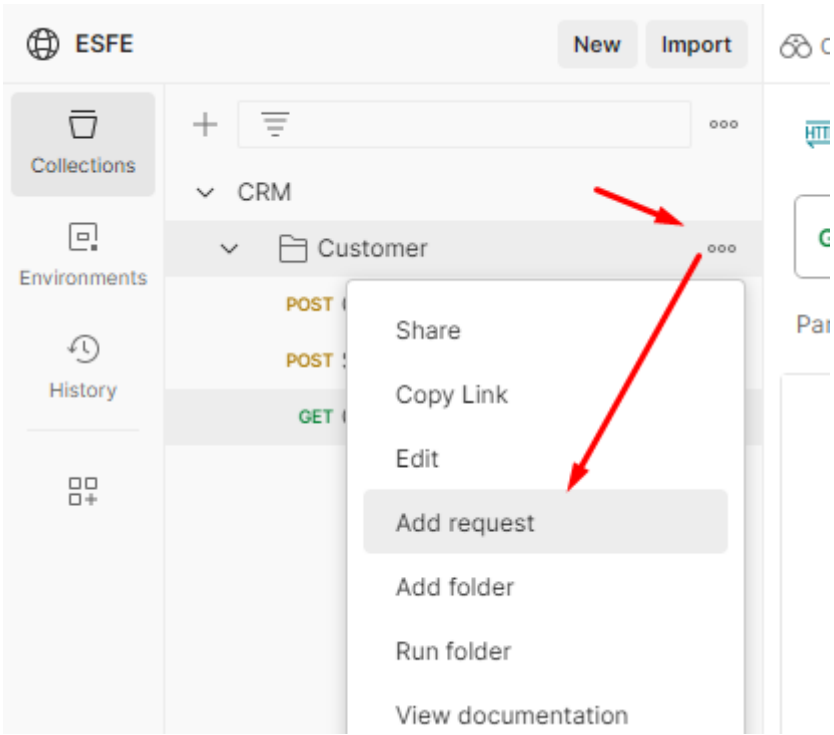

CRM / Customer / GetId
Save
GET
{{url}}/customer/{{clientid}}
Send
Params
Authorization
Headers (7)
Body
Pre-request Script
Tests
Settings
Cookies
1 pm.test("Status code is 200", function () {
2 pm.response.to.have.status(200);
3 });
Write scripts with Postbot
Test scripts are written in JavaScript, and are run after the response is received.
Learn more about tests scripts

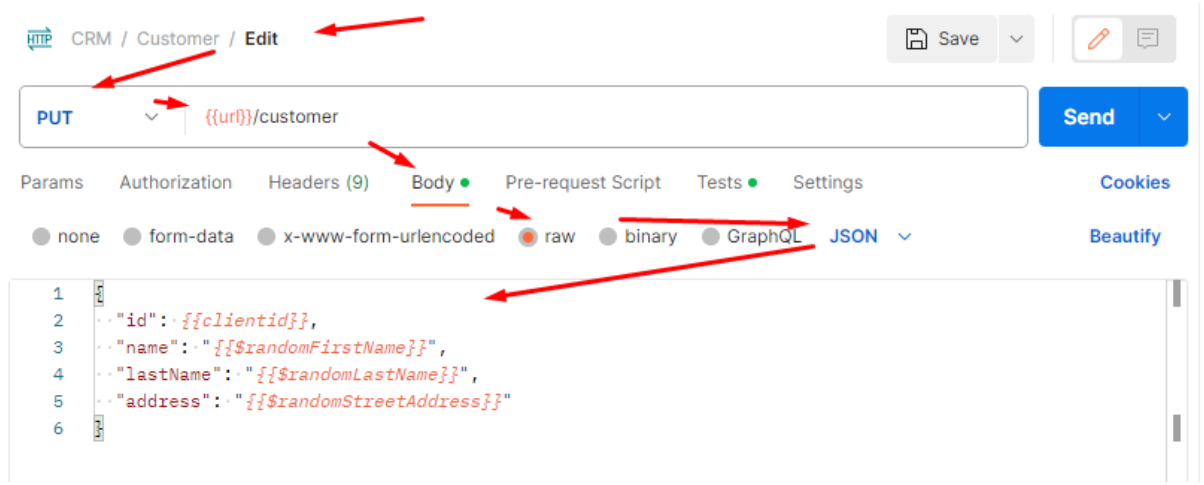

CRM / Customer / GetId
Save
GET
{{url}}/customer/{{clientid}}
Send
Params
Authorization
Headers (7)
Body
Pre-request Script
Tests
Settings
Cookies
1 pm.test("Status code is 200", function () {
2 pm.response.to.have.status(200);
3 });
Write scripts with Postbot
Test scripts are written in JavaScript, and are run after the response is received.
Learn more about tests scripts
Snippets
Body
Cookies (1)
Headers (4)
Test Results (1/1)
200 OK
18 ms
226 B
Save as Example
Pretty
Raw
Preview
Visualize
JSON
1 {
2 "id": 3002,
3 "name": "Patrick",
4 "lastName": "Thiel",
5 "address": "7975 Effertz Haven"
6 }


CRM / Customer / GetId
Save
GET
{{url}}/customer/{{clientid}}
Send
Params
Authorization
Headers (7)
Body
Pre-request Script
Tests
Settings
Cookies
1 pm.test("Status code is 200", function () {
2 pm.response.to.have.status(200);
3 });
Write scripts with Postbot
Test scripts are written in JavaScript, and are run after the response is received.
Learn more about tests scripts
Snippets
Body
Cookies (1)
Headers (4)
Test Results (1/1)
200 OK
18 ms
226 B
Save as Example
All
Passed
Skipped
Failed
PASS
Status code is 200

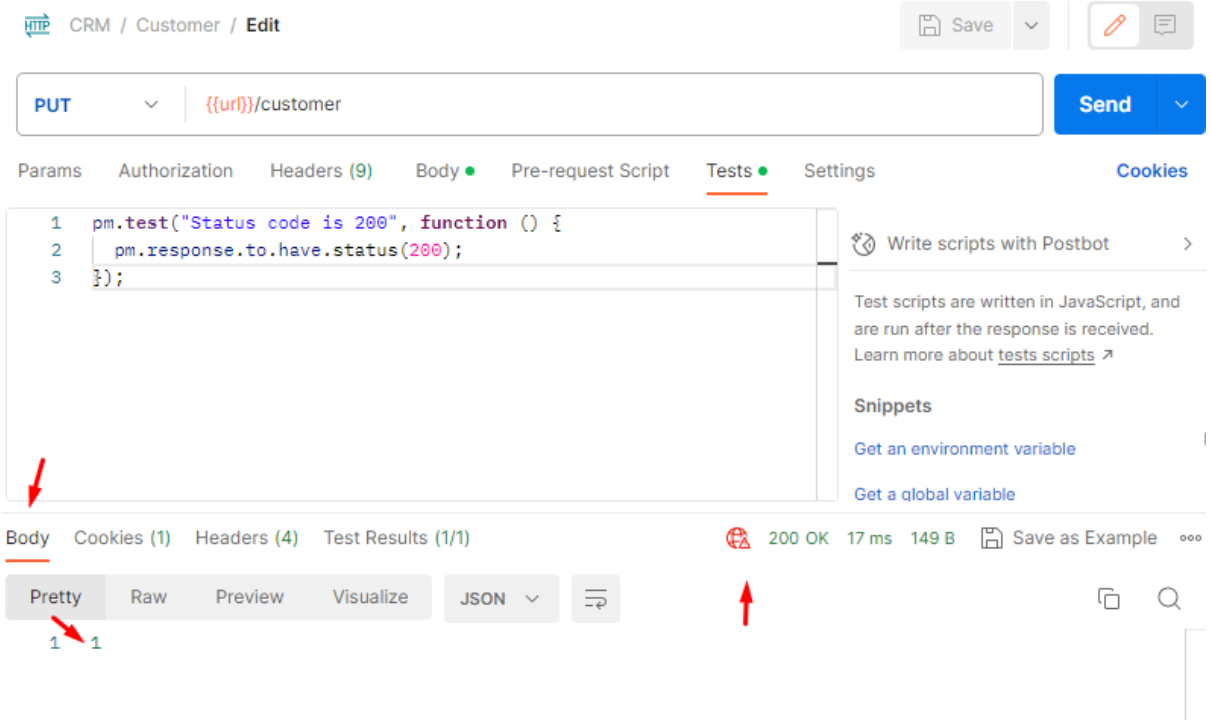


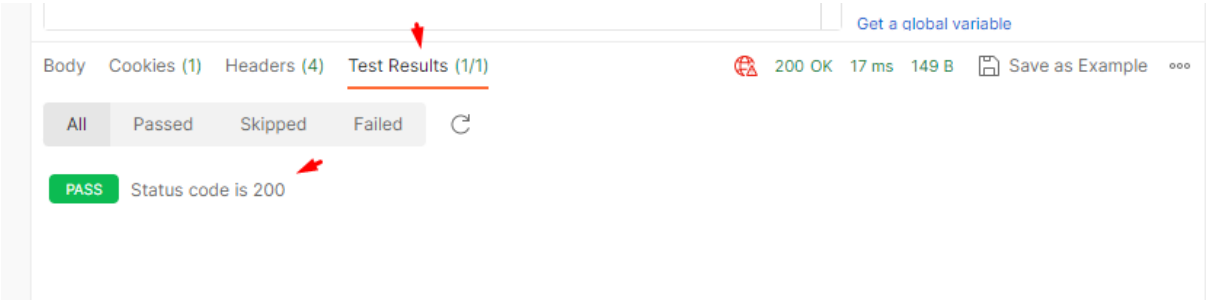
Ahora vamos a crear el endpoint para editar.

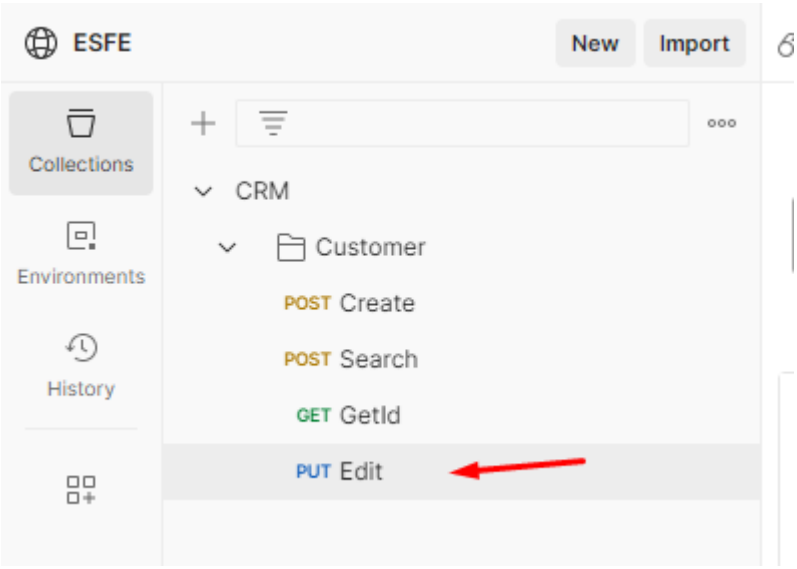



CRM / Customer / Edit
Save
PUT
{{url}}/customer
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"id": {{clientid}},
"name": "{{$randomFirstName}}",
"lastName": "{{$randomLastName}}",
"address": "{{$randomStreetAddress}}"
}

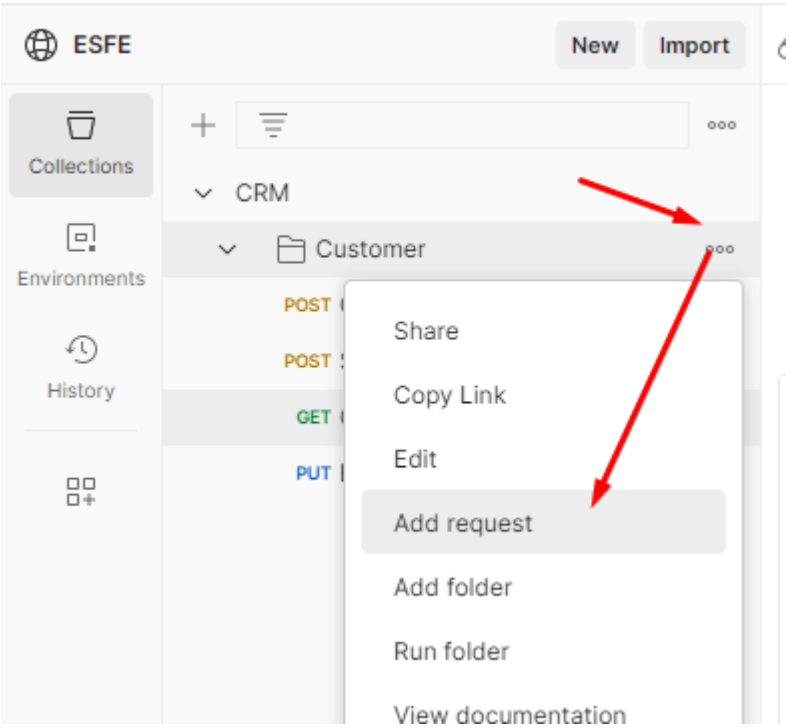

CRM / Customer / Edit
Save
PUT
{{url}}/customer
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
pm.test("Status code is 200", function () {
pm.response.to.have.status(200);
});
Write scripts with Postbot
Test scripts are written in JavaScript, and are run after the response is received.
Learn more about tests scripts
Snippets
Get an environment variable

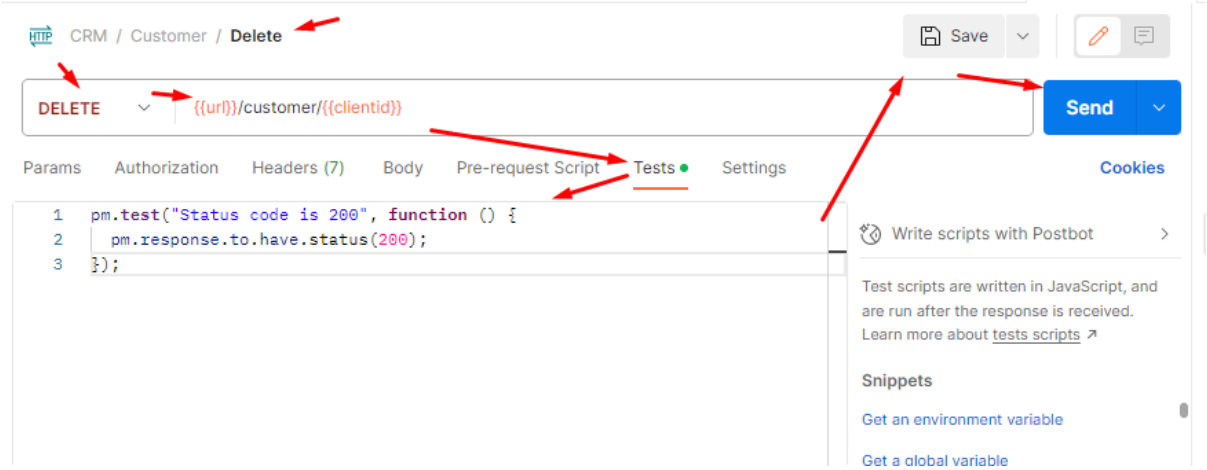

CRM / Customer / Edit
Save
PUT
{{url}}/customer
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
pm.test("Status code is 200", function () {
pm.response.to.have.status(200);
});
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Snippets
Get an environment variable
Get a global variable
Body
Cookies (1)
Headers (4)
Test Results (1/1)
200 OK
17 ms
149 B
Save as Example
Pretty
Raw
Preview
Visualize
JSON
1 1

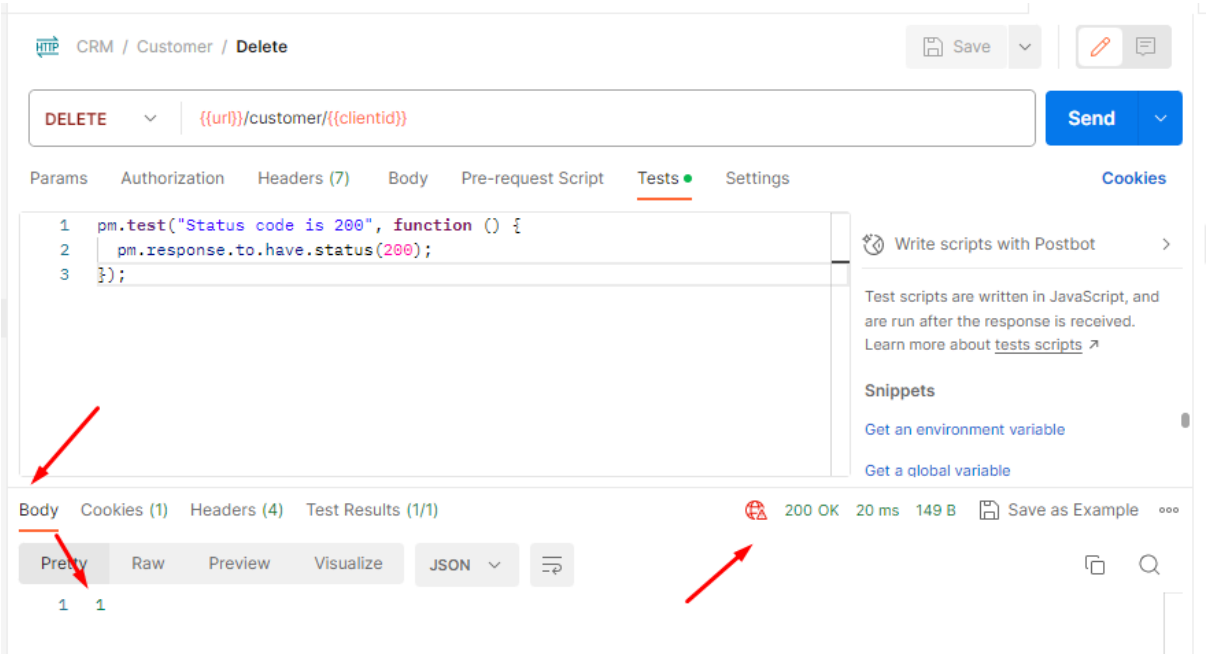


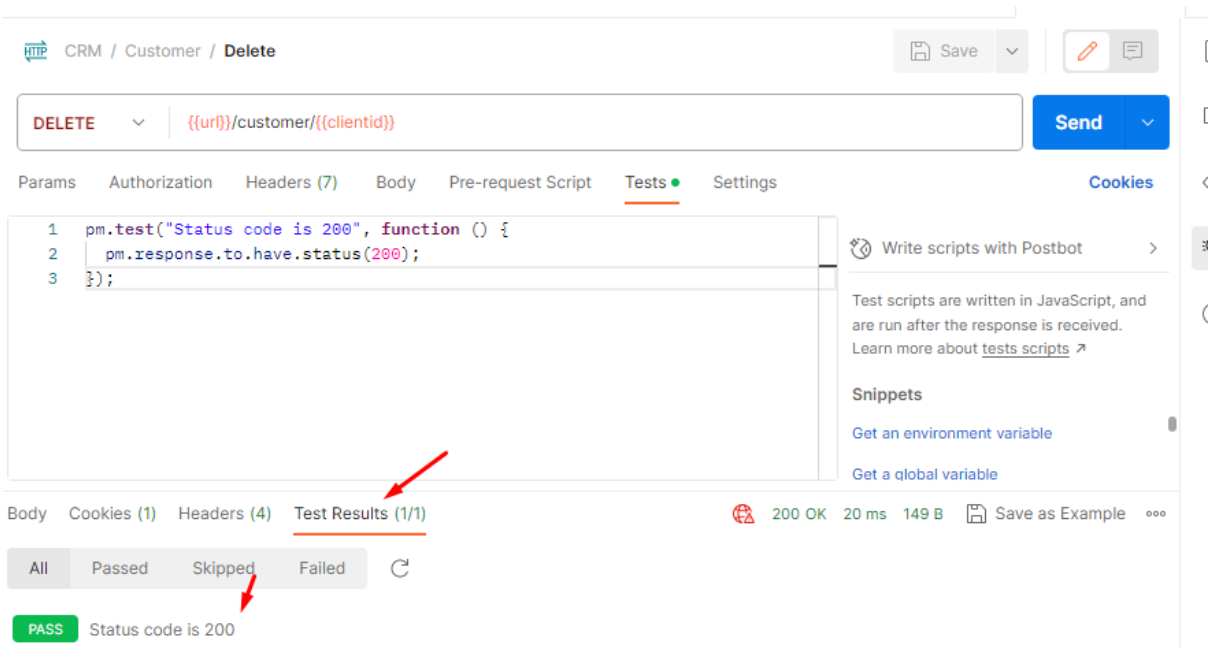


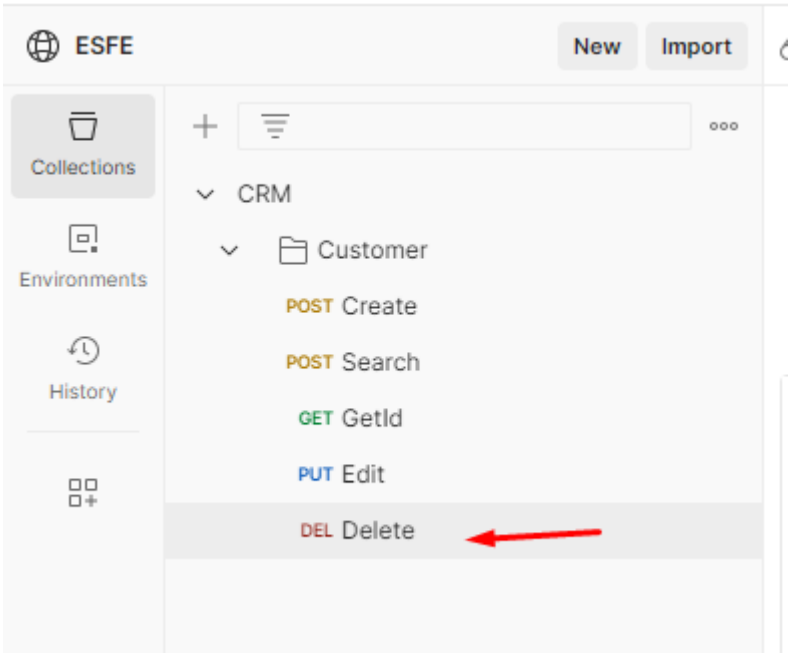
Ahora vamos a crear el endpoint para eliminar.



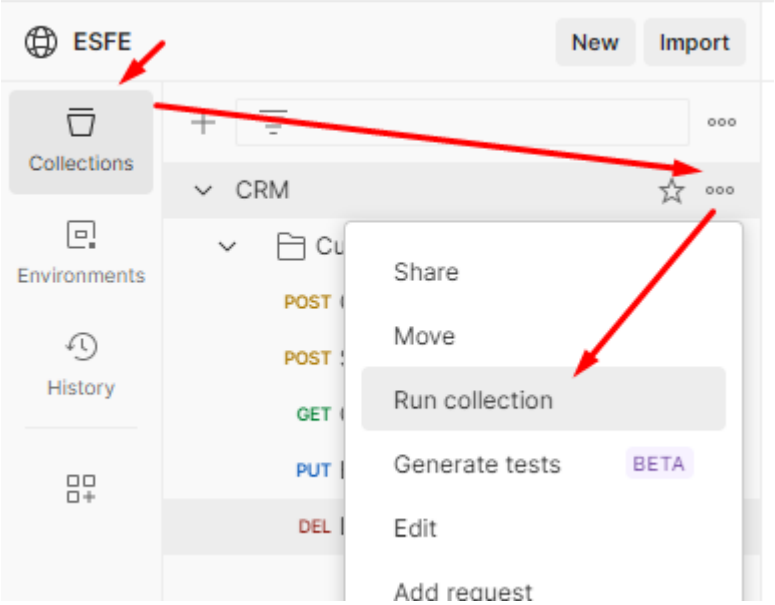

CRM / Customer / Delete
Save
DELETE
{{url}}/customer/{{clientid}}
Send
Params
Authorization
Headers (7)
Body
Pre-request Script
Tests
Settings
Cookies
1 pm.test("Status code is 200", function () {
2 pm.response.to.have.status(200);
3 });
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Get a global variable

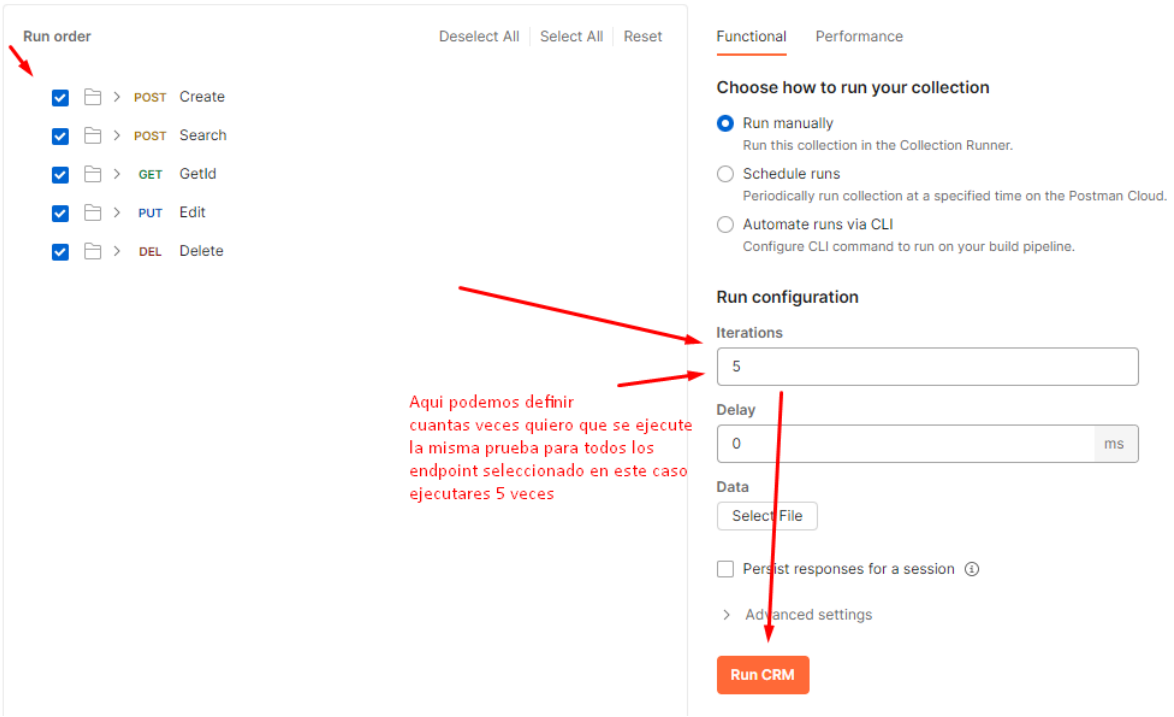

CRM / Customer / Delete
Save
DELETE
{{url}}/customer/{{clientid}}
Send
Params
Authorization
Headers (7)
Body
Pre-request Script
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Cookies (1)
Headers (4)
Test Results (1/1)
200 OK
20 ms
149 B
Save as Example
Pretty
Raw
Preview
Visualize
JSON
1 1



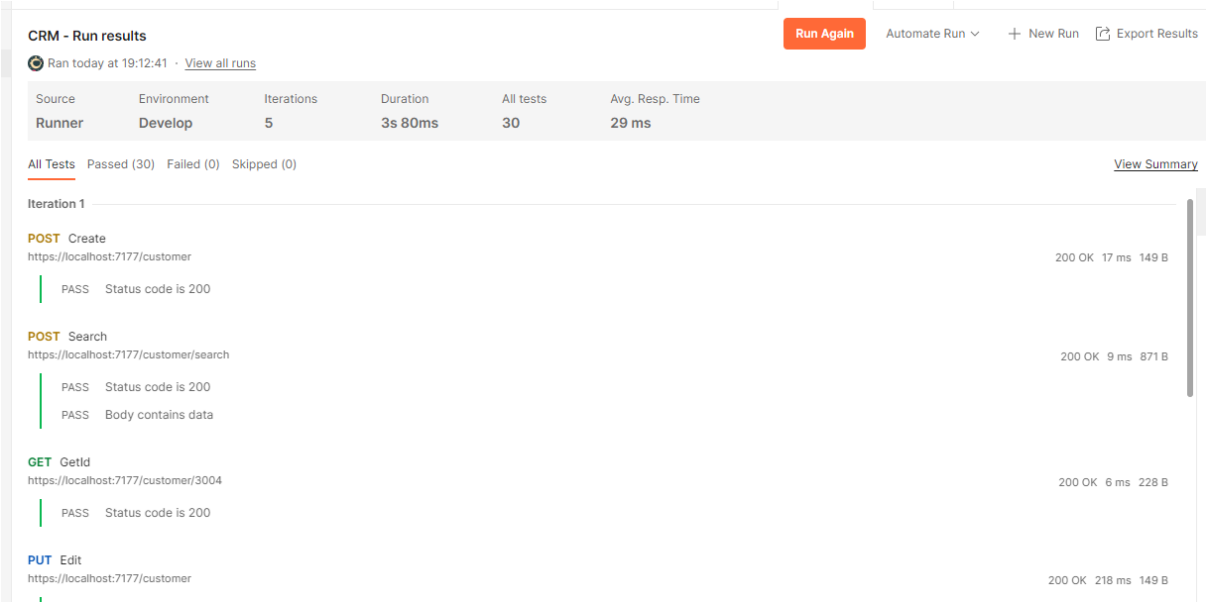


Ahora aprenderemos cómo ejecutar todos los endpoints de una sola vez que se encuentran en mi collection CRM y ver un análisis global de las pruebas realizadas.

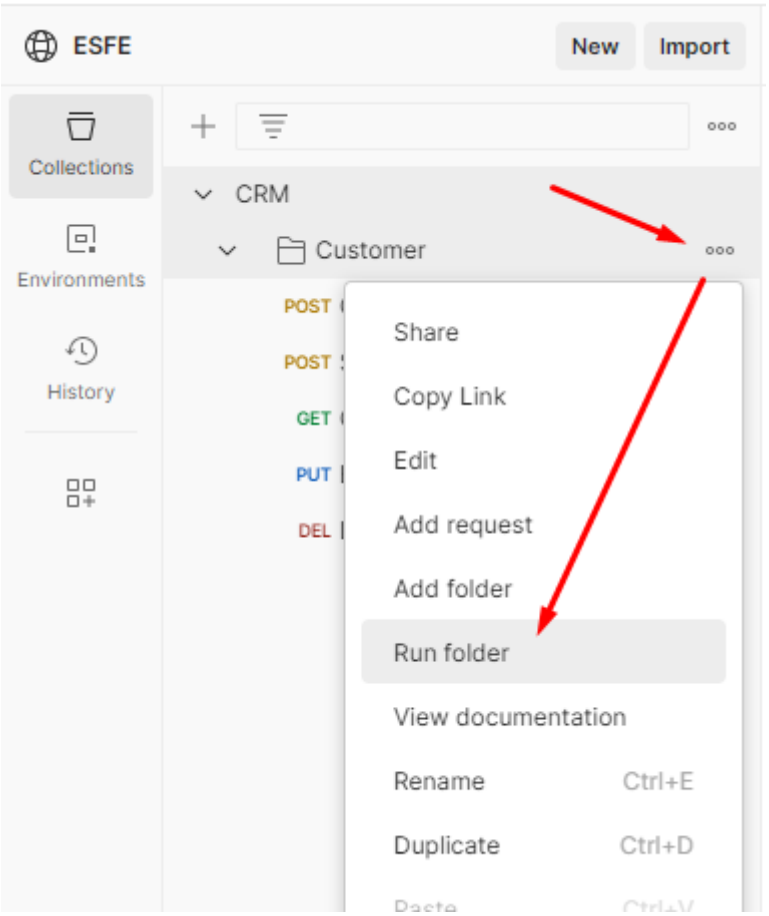


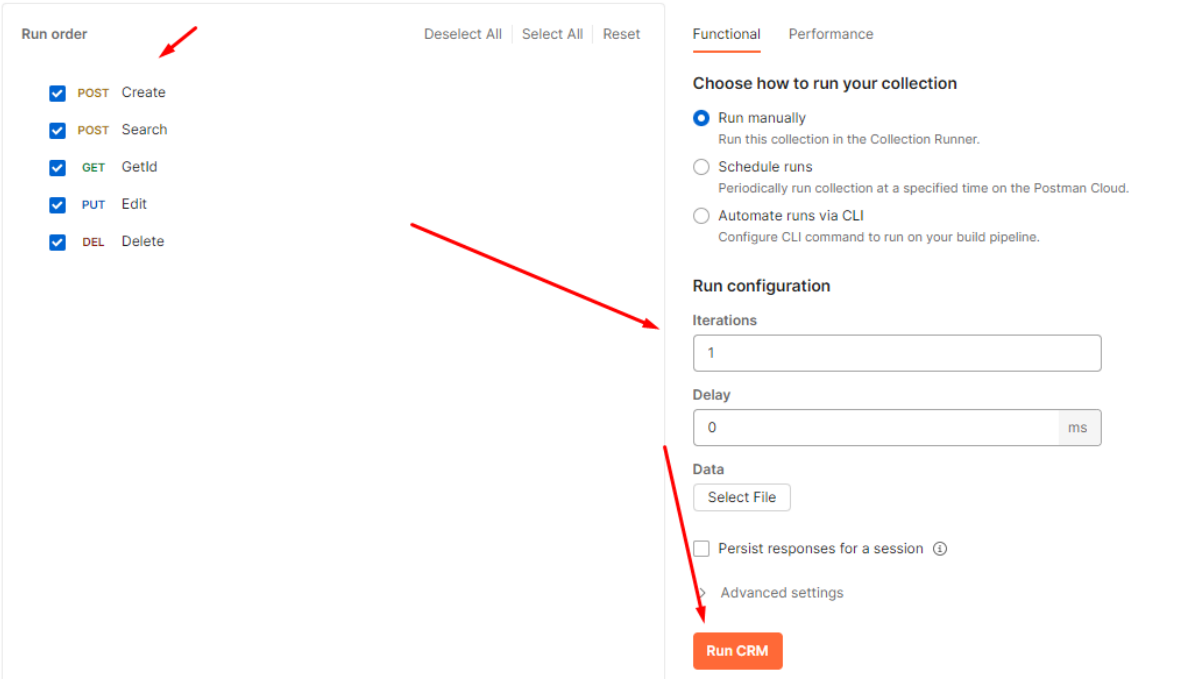


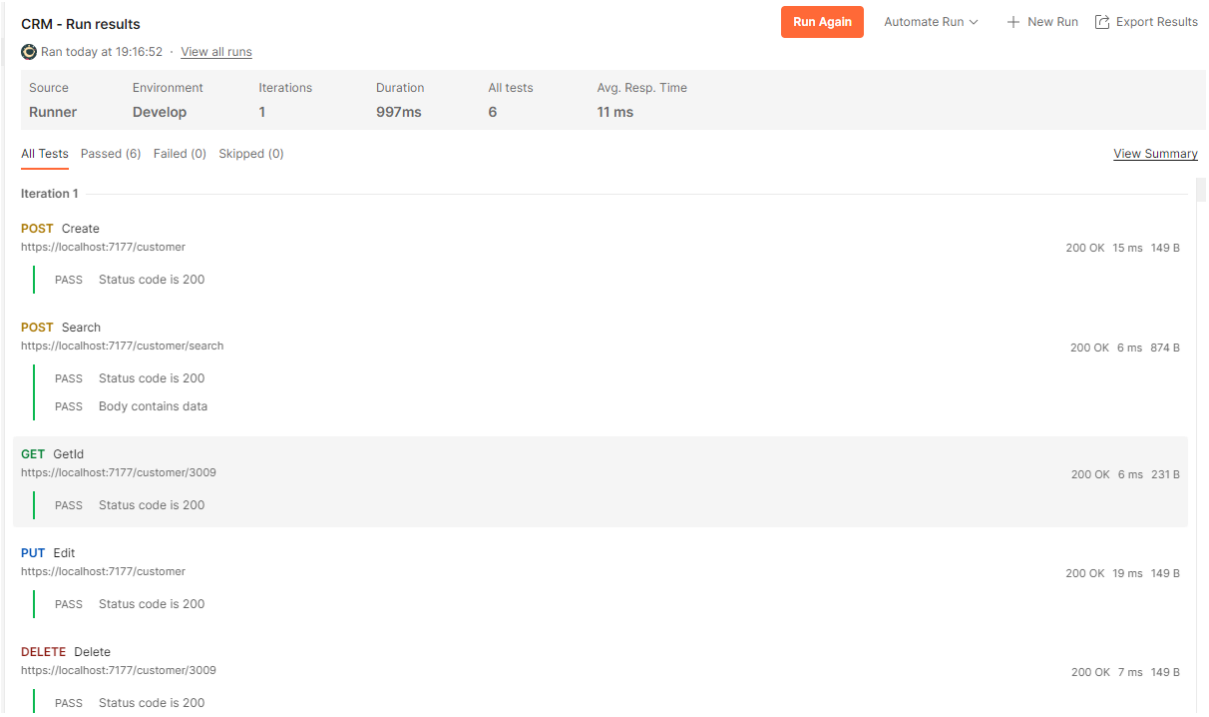
Después de ejecutar, se mostrará un análisis de las pruebas realizadas, lo que nos permitirá ver de manera general cuáles pruebas tuvieron éxito y cuáles fallaron.



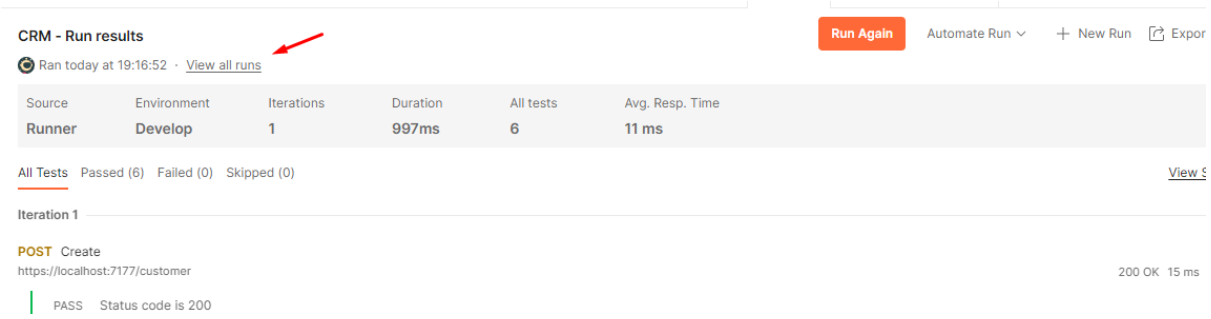
También se puede ejecutar los test por folder

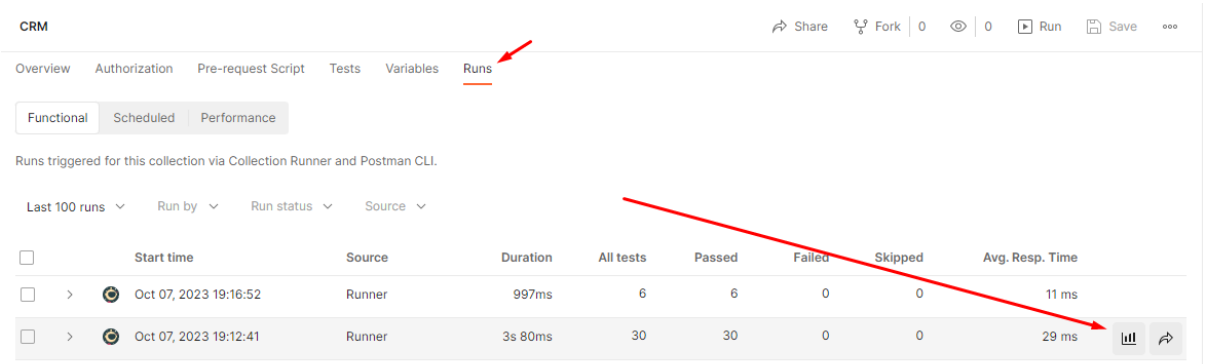


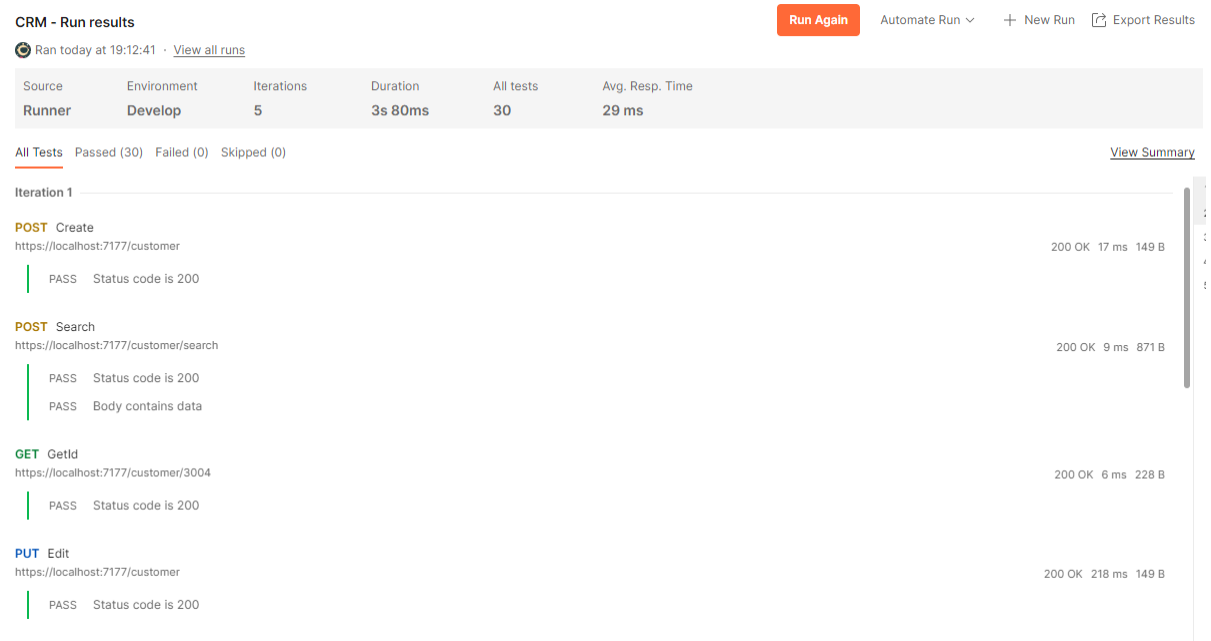




También podemos ver un historial de cada una de las pruebas realizadas







Ejercicio:
Testeo en postman al ejercicio construido en el módulo anterior en la guia 5 .net

Indicaciones:

- Documentar el caso de prueba en azure devops
- Testear cada endpoint de la api debe de quedar documentada en su cuenta de postman.